

Human-like autonomy emerges from self-play and a pinch of human data

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Abstract: Self-play reinforcement learning has recently taken the wheel in autonomous driving. The approach uses cheap, large-scale simulation to substitute expensive, large-scale human demonstration data. While cost-effective, agents trained through self-play often behave unpredictably around unseen policies and fail to coordinate with human drivers due to incompatible driving styles. Previous works attempt to mitigate this behavioral incompatibility through reward engineering and domain randomization, which is labor-intensive and brittle. We build on the best of both approaches and train driving policies by training on over 60 years of self-play simulation with a minimal reward function and just 30 minutes of human demonstrations. Our agents achieve near-zero-shot coordination with real human driving trajectories using $125,000 \times$ less human data than imitation learning baselines, at $1/25$ the number of GPU hours. The solution space of unconstrained driving policies is large, but this small pinch of human data is sufficient to act as a regularizer towards ones that fit the road. Together, this suggests that human-compatible agents can be trained with far less human demonstration data than is currently assumed.

Keywords: Zero-Shot Coordination, Self-play Reinforcement Learning, Imitation Learning, Autonomous Driving

1 Introduction

Self-play reinforcement learning (RL) has produced superhuman agents in strategic games [1, 2, 3] and, more recently, has shown promise in real-world domains including autonomous driving [4, 5, 6, 7] and robotic manipulation [8]. The approach elegantly sidesteps a central difficulty in multi-agent learning - how to model the opponent - through the following idea: the *agent’s opponent is a copy of itself*. The appeal here is that as the agent improves, so does its co-player. This gives rise to an automatically evolving curriculum [9] that takes behavior from random play to optimal behavior entirely through synthetic simulated experience.

In zero-sum games, this mechanism with a sparse measure for success (e.g., +1 when winning a game of chess) is enough to produce strong play against arbitrary opponents. Many real-world settings, however, are not zero-sum. Driving, for instance, is a mixed-motive game: each agent has *individual objectives* (reaching a destination safely) but must also *coordinate* with other road users by adhering to shared norms, expectations, and conventions. Self-play RL with only a high-level objective for success provides no guarantees of such alignment; agents may converge to effective but “alien” strategies that are incompatible with human partners [10]. Concretely, an agent trained to “reach a destination safely,” may very well learn to do so in reverse, sideways, or on the wrong side of the road if such constraints are not specified in the reward.

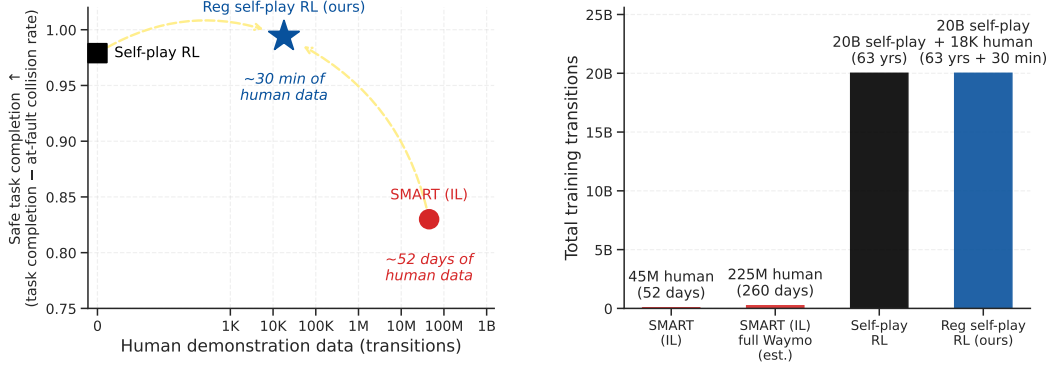


Figure 1: **Regularized self-play RL achieves better safe coordination with human proxies using a fraction of the human demonstration data.** *Left:* safe task completion versus human demonstration data per method. Safe task completion is defined as task completion rate $-$ at-fault collision rate, evaluated by placing each agent in scenes populated by human-replay proxies (i.e. other agents replaying recorded human trajectories); it jointly measures task effectiveness and ability and human-compatibility. Adding ~ 30 minutes of human demonstrations as an RL regularizer (star, ours; safe task completion $100\% - 0.06\% = 0.994$) outperforms both the unregularized self-play baseline (square; $100\% - 2.1\% = 0.979$) and SMART (circle; $84.6\% - 1.6\% = 0.830$), an imitation-learning model trained on the full Waymo dataset. Yellow arrows indicate the improvement over each baseline. *Right:* total training transitions per method. Our method and the self-play RL baseline each train on 20B transitions (~ 63 years of simulated driving at 10 Hz), almost entirely on cheap synthetic self-play data. SMART-tiny-CLSFT trains on 45M between human demonstration transitions (only including the SDC trajectory; ~ 52 days) or 225M transitions (using all agent trajectories per scene ~ 7 months) is shown for reference. Experience to human time estimation details in Appendix F.

Previous works have addressed this misalignment in two ways. One line of work involves *manual reward engineering*, where reward terms are added iteratively until the desired behavior and conventions emerge [5, 11]. While effective, this strategy is labor-intensive by nature, domain-specific, and brittle since it is not trivial to figure out what reward will produce the desired human-like behavior [12]. A good example is GIGAFLow [5], which required nine individually tuned reward terms and several other domain randomization tricks to produce naturalistic driving behavior. On the other side of the spectrum we have *Imitation Learning* [13, 14, 15]. IL directly imitates human demonstrations to avoid having to define a reward function altogether. However, robustness requires broad behavioral coverage, so these approaches typically require large quantities of human demonstrations to work well [16].

We take a different approach, grounded in a practical observation about the changing cost structure of experience generation. Modern RL frameworks and simulation infrastructure can generate between 300K and 20M environment steps *per second* on a single consumer-grade GPU [17, 18][PufferLib], making synthetic experience effectively limitless. Human demonstrations, by contrast, require manual collection and remain difficult to scale. This suggests a natural role for human demonstration data in games with the aforementioned properties: not as the primary source of training signal, but as a lightweight anchor that steers the policy away from effective yet behaviorally alien strategies. Indeed, regularizing self-play RL toward such an anchor has shown promise in producing human-compatible agents in Diplomacy [19, 20] and driving [21, 22, 7], yet *how much* demonstration data is required to reach human compatibility remains, to our knowledge, unexamined.

We present the first systematic study of this question. We regularize self-play PPO toward a behavioral cloning anchor trained on varying amounts of human demonstrations from the Waymo Open Motion Dataset [23, WOMD], and find that the answer is: surprisingly little. Human-compatible driving agents can be learned from as little as **30 minutes to 3 hours of human demonstrations**, roughly **0.04 to 0.25%** of the human demonstration data used to train state-of-the-art imitation learning

61 models, while using 1/25 of the GPU hours times faster. More broadly, the present results suggest
62 that **wherever simulation is cheap, collecting a small amount of human demonstrations may**
63 **suffice** to learn agents that are compatible with human play.

64 Our contributions are:

- 65 • We show that regularizing self-play RL toward a behavioral cloning anchor achieves im-
66 proved coordination with human drivers where pure self-play fails, without reward engineer-
67 ing and with 30 minutes to 3 hours of human demonstration data when combined with 60
68 years of synthetic self-play generated experience.
- 69 • We present the first systematic study of human data scaling under regularized self-play RL
70 in the application domain of autonomous driving.
- 71 • We open-source the full codebase; each training run is reproducible in seven hours on a
72 single consumer-class GPU.

73 2 Related Work

74 **Imitation learning for autonomous driving.** The generation of driving policies is a fundamental
75 challenge across end-to-end autonomous driving [24, 25, 26, 27], multi-agent trajectory prediction
76 [28], and reactive traffic simulation [29, 30]. Driven by the widespread availability of large-scale
77 human driving datasets [31, 23, 32], imitation learning has become the dominant approach across all
78 these domains [33]. Under this imitation learning paradigm, a broad spectrum of methodologies has
79 emerged to fit models to historical data, ranging from marginal [34, 35, 36] and joint [37, 38, 39, 40]
80 forecasting to autoregressive sequence modeling of tokenized trajectories [41, 15, 42] and continuous
81 distribution learning via diffusion and promptable world models [43, 44, 45, 46, 47]. While these
82 generative approaches yield diverse open-loop behaviors, they are fundamentally constrained by
83 the scale of human data required and frequently suffer from compounding covariate shift in closed-
84 loop deployment [16]. To mitigate these shifts, recent hybrid approaches integrate reinforcement
85 learning [48, 49, 50], yet they typically still rely on extensive human demonstrations as their primary
86 optimization signal. Our approach systematically inverts this balance: rather than depending on
87 human driving logs as the core supervisor, we utilize synthetic, multi-agent RL self-play as the
88 primary engine for discovering robust interactive behaviors, retaining a remarkably small human
89 dataset strictly as a behavioral anchor to ensure conformity to realistic traffic norms.

90 **Self-play reinforcement learning in games.** Self-play reinforcement learning has produced super-
91 human agents in games from Go and Chess [1, 51] to StarCraft II [52] and Stratego [2], all without
92 human data. Superhuman play is not the same as human-compatible play, however. Many games
93 admit multiple equilibria, and self-play need not converge to equilibria that are compatible with
94 human partners [10, 53]. The failure has been shown in cooperative games such as Hanabi [54]
95 and Diplomacy [10], where self-play agents develop internally consistent conventions that transfer
96 poorly to human partners. The cause is reward underspecification: when the reward is defined as a
97 score to maximize, there are often many ways to achieve it. In other words, the solution space is
98 large. Previous work attempts to resolve this by sharing the reward by hand [5, 11]. For instance,
99 GIGAFLOW [5] demonstrates that reward engineering and domain randomization can produce
100 naturalistic behavior at scale, at the cost of nine individually tuned reward terms. We avoid reward
101 engineering entirely. A small amount of human data serves as a behavioral anchor, and self-play does
102 the rest. This reduces a labor-intensive design problem to a one-hour data collection.

103 **Human-regularized self-play reinforcement learning and search.** One alternative to reward
104 engineering is to regularize self-play toward a human anchor policy. This idea has been explored
105 in Diplomacy, where KL regularization toward a human prior during both search and learning
106 produced agents that coordinate more effectively with human partners [19, 20]. Jacob et al. [55]
107 study KL-regularized search more broadly and show that it recovers human-like play across several
108 games. In autonomous driving, Cornelisse and Vinitzky [21] show that regularizing PPO toward a

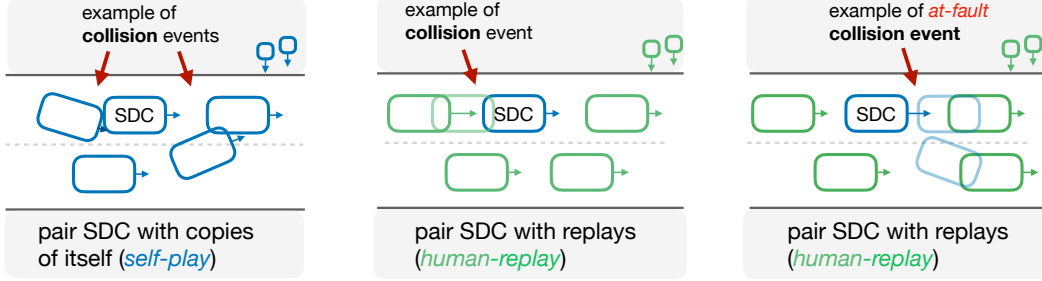


Figure 2: **Evaluation settings.** Self-play (left) and human-replay (center, right). Red arrows mark collisions. Rectangles are vehicles; squares are pedestrians. In human-replay, some collisions are effectively unavoidable: replay agents follow their logged trajectories and can drive into the controlled SDC from behind. We therefore distinguish between *collisions* (any contact) and *at-fault collisions* (contact caused by the controlled agent, following the NAVSIM benchmark [58]).

behavioral cloning anchor produces more human-compatible driving agents on a small scale. More recently, Chang et al. [7] show that KL-regularized self-play can train human-like driving policies from scratch using SMART [42] as the behavioral anchor, a large tokenized model trained on the full 500,000-scenario Waymo dataset. [22] show that regularized self-play RL effectively adapts policies to new cities, without using human demonstrations from that city. In sum, the effectiveness of regularized self-play RL has been shown. What remains unexamined is the scaling law that governs it: how performance depends on the amount and quality of the human anchor data. The question matters now in a way it did not before. Simulation throughput has grown by orders of magnitude in recent years, shifting the bottleneck from synthetic experience to human data.

3 Method

3.1 Evaluation and Metrics

Evaluation. A human-compatible agent should *blend in* with human drivers. Since on-road deployment is out of scope, we approximate the human population with logged human trajectories and evaluate in three settings, illustrated in Figure 2:

- **Self-play.** All agents are controlled by the same trained policy in a decentralized manner.
- **Human-replay.** Only the self-driving car (SDC) is controlled by the policy; all other agents follow their logged trajectories.
- **IDM.** The SDC is controlled by the policy; all other agents follow the Intelligent Driver Model [56], tracking a precomputed lane-center path for lateral control and using IDM longitudinal accelerations to maintain a safe gap to the lead vehicle, as in [57].

A human-compatible agent should score well across all three: it should reach its goal without collisions or off-road events whether its co-players are copies of itself, reactive rule-based drivers, or human drivers it has never trained against. The three settings probe complementary failure modes. Human-replay is the strictest test: because logged trajectories cannot react, the SDC alone is responsible for avoiding collisions, so the setting stress-tests whether the policy has internalized human driving conventions rather than coordinating on idiosyncratic self-play conventions. IDM sits between self-play and human-replay in difficulty; co-players are no longer the policy itself (so self-play-specific conventions do not help), but they are reactive, so the SDC is not solely responsible for averting every collision.

Metrics and ceiling. Our primary metric is the **score**: an agent scores 1 if it completes the task of driving to a goal destination without colliding or going off-road, and 0 otherwise. To diagnose

failure modes, we separately report **collision rate**, **at-fault collision rate**, **off-road rate**, and **route progress**. An ideal agent should score well with its own population as well as the human-replay population. Score-based metrics capture whether agents complete their task safely, but not whether their behavior looks human. We therefore also report **distributional realism** using the Waymo Open assess whether agents complete their tasks safely, but not whether their behavior appears of simulated and logged trajectories across kinematic, interactive, and map-based features. All metrics are reported on held-out test scenarios unless stated otherwise; full definitions are given in Appendix D.2.

3.2 Simulation Environment

We train policies $\pi(a_t^i | o_t^i)$ in PufferDrive 2.0 [18], a batched driving simulator that reaches 300k SPS on a single consumer-grade GPU. Environments are initialized from the Waymo Open Motion Dataset [23]: each 9-second scenario provides a roadgraph, a variable set of agents (cars, cyclists, pedestrians) up to $N = 32$, and per-agent initial poses and goals drawn from the logged trajectories. Each agent receives a partial, decentralized, ego-frame observation consisting of its own state, the $N - 1$ closest neighbors within 50 m, and up to 128 nearby road segments.

We avoid reward shaping and use a sparse reward driven only by high-level outcomes: +1 for reaching the goal position, -1 for collision or going off-road, and 0 otherwise. This way, human-like behavior emerges from regularization rather than from a hand-tuned reward. Episodes are truncated once all agents reach their destinations. Full details are given in Appendix A.

3.3 Regularized Self-Play Reinforcement Learning

Our method has two stages: we first train a behavioral cloning (BC) anchor on human demonstrations, then use it as a regularizer during self-play RL. Note that the RL policy is learned from scratch rather than fine-tuned from the anchor.

Step 1: Train the anchor policy. To study how the amount of human demonstration data affects downstream performance, we train a set of anchor policies on nested subsets of the demonstrations. Given the full dataset $\mathcal{D} = \{(o_t^i, a_t^i)\}_{i=1}^{T \cdot K}$ of observation-action pairs from K scenarios of T timesteps each, we sample subsets $\mathcal{D}_n \subset \mathcal{D}$ corresponding to n scenarios, yielding roughly $\{10 \text{ min}, 30 \text{ min}, 3 \text{ h}, 30 \text{ h}\}$ of human driving data. Each anchor τ_n is trained by minimizing the negative log-likelihood of the logged actions:

$$\tau_n = \arg \min_{\tau \in \mathcal{T}} \sum_{(o_t^i, a_t^i) \in \mathcal{D}_n} -\log \tau(a_t^i | o_t^i), \quad (1)$$

Figure 10 illustrates the data collection process in three scenarios. Since we have enough scenarios, we only use the SDC trajectory from each scenario, as it is typically the highest-quality trajectory in the scene. Each τ_n is then frozen and used as the regularizer during the subsequent self-play stage. See Appendix A.4 for the full details.

Step 2: Regularized self-play RL. We train a new policy π_θ from scratch using PPO [59] in self-play, with each anchor τ_n acting as a behavioral regularizer. Concretely, we add a KL penalty from the anchor to the policy to the standard PPO objective:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{PPO}}(\theta) + \lambda \mathbb{E}_{o \sim \rho_{\pi_\theta}} [D_{\text{KL}}(\tau_n(\cdot | o) \parallel \pi_\theta(\cdot | o))], \quad (2)$$

where ρ_{π_θ} is the state distribution induced by the on-policy rollouts and $\lambda \geq 0$ controls the regularization strength. The KL term pulls π_θ toward the anchor’s action distribution on states the policy actually visits, rather than on the distribution of states in $\mathcal{D}_n$. We use scenario metadata to initialize worlds during self-play (Appendix A.1). The full training objective and hyperparameters are given in Appendix B.

4 Experiments

This section summarizes the key results. Additional details and experiments are reported in the appendices, including details on datasets, network architectures, hyperparameters, and environment settings. The appendices also report ablations.

The sections are structured to answer the following questions:

1. *Scaling laws for human demonstration data*: How much human demonstration data is needed for strong performance in both the self-play and human-replay settings? How do these scaling laws compare to those of the state-of-the-art imitation learning baseline? (Section 4.1)
2. *The role of metadata and scenario diversity*: Driving datasets such as WOMB and NuPlan provide scenario metadata—road graphs and initial agent positions—that grounds simulation at a fraction of the cost of collecting human demonstrations. How does the number of training scenarios (maps) used for self-play influence agent performance? (Section 4.2)
3. *Behavior and safety analysis*: Human-likeness in driving is not easily captured by any single metric. We conduct a full behavioral analysis comparing regularized self-play agents against their unregularized counterparts and the imitation learning baseline. (Section 4.3). Videos to show the difference in agent behavior are provided on an anonymous site: <https://sites.google.com/view/anonymous-human-like-autonomy>

4.1 Scaling Laws for Human Demonstrations

How much human demonstration data does regularized self-play need, and how does this compare to pure imitation learning? The second question matters because any apparent data efficiency on our side could simply reflect the homogeneity of the Waymo Open Dataset rather than an actual property of the method.

To answer this, we benchmark against SMART-mini [42, 50] (—●—), the state-of-the-art imitation-learning approach in this field. We train SMART on the same nested subsets of human demonstrations as regularized self-play (—●—). As a second baseline, we include vanilla self-play RL with the same minimal reward function (Appendix B.2). Both regularized and unregularized agents are trained for 20 Billion steps of experience, where each run takes 15 hours to complete on a single NVIDIA 5090 GPU (390k SPS). Figure 3 shows the results visually (See Figure 17 for log scale); Table 1 reports the numbers and additional metrics.

Regularized self-play RL outperforms SMART across all conditions at less than 0.25% of the human data budget. Regularized self-play outperforms SMART-mini across every data bin and metric we evaluate. Plain self-play RL achieves a human-replay at-fault collision rate of 2.1% (---). Adding just 30 minutes to 3 hours of human demonstrations (—●—) drives this down to 0.6%. SMART-mini (—●—), trained on 52 days of human data, reaches only 1.6%, which is two to three times higher than regularized self-play RL using less than 0.25% of that data budget. Regularized self-play also achieves near-perfect task completion, whereas SMART agents achieve 90% route progress at best. This gap can be explained by training experience: self-play RL consumes 20 billion transitions versus 225 million for SMART (Figure 1). The difference is that most of that experience is cheap simulated experience rather than collected from human drivers.

The lower panel of Figure 3 quantifies the *relative* performance improvements of regularized self-play RL to SMART. At 30 minutes of human data, regularized self-play RL reduces the self-play collision rate by $46\times$, the human-replay at-fault collision rate by $11\times$, and improves route progress by $1.3\times$ relative to SMART, a strict improvement across every metric, as further supported in Table 1.

Regularized self-play agents perform well across partner populations. Regularized self-play agents perform well whether paired with their own policy or with held-out human trajectories, maintaining low collision rates in both settings. SMART, even with 52 days of training data, incurs a

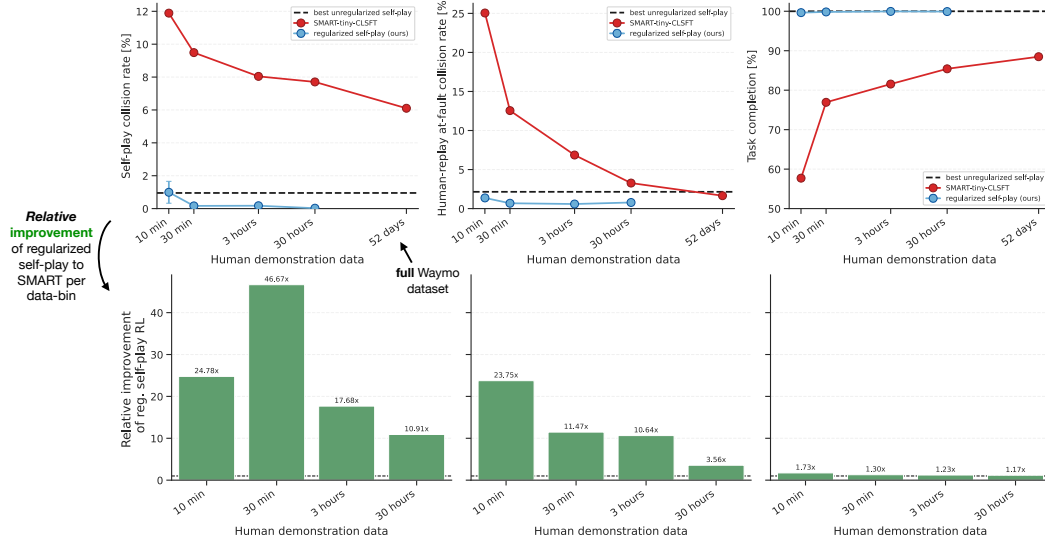


Figure 3: **Scaling laws for human demonstration data.** Performance of *regularized* self-play RL (—●—) and SMART with CAT-K closed-loop fine-tuning (—●—) as a function of the amount of human demonstration data used for training, evaluated in self-play and against human replays. All policies are evaluated on the same 10k held-out scenarios from the Waymo Open validation set [23]. Regularized self-play saturates at roughly three hours of human data, whereas SMART continues to improve across the full WOMD training set. Self-play RL (---) is shown as a horizontal line, since it uses no human demonstrations. The horizontal axis is semi-logarithmic.

227 6% self-play collision rate, which is much higher than the 0.0–1.0% achieved by regularized self-play
 228 RL under comparable human demonstration budgets.

Table 1: Performance vs. amount of human demonstration data for the best trained policies on 10,000 held out randomly sampled scenarios. Top-3 values per column are highlighted (best, 2nd, 3rd); best value additionally in bold. The unregularized self-play row uses no human demonstrations.

Human demos used	Method	Self-play (test)			Human-replay (test)				
		Coll. (%) ↓	Off-road (%) ↓	Route prog. (%) ↑	Score ↑	Coll. (%) ↓	At-fault (%) ↓	Off-road (%) ↓	Route prog. (%) ↑
10 min	SMART	11.9	55.8	84.5	0.246	32.0	25.0	18.6	57.7
30 min	SMART	9.5	55.4	85.8	0.379	17.9	12.5	16.8	76.9
3 hours	SMART	8.0	53.6	86.2	0.518	11.4	6.9	4.5	81.5
52 days	SMART	6.1	53.5	91.7	0.654	4.4	1.6	1.1	88.5
—	unreg. self-play	1.0 ± 0.4	0.2 ± 0.2	99.9 ± 0.1	0.967 ± 0.006	2.7 ± 0.5	2.1 ± 0.5	0.6 ± 0.2	100.0 ± 0.0
10 min	reg. self-play (ours)	1.0 ± 0.7	0.3 ± 0.2	99.0 ± 0.4	0.941 ± 0.007	3.9 ± 0.6	1.4 ± 0.4	1.4 ± 0.4	99.6 ± 0.2
30 min	reg. self-play (ours)	0.2 ± 0.1	0.5 ± 0.2	99.3 ± 0.3	0.968 ± 0.006	2.0 ± 0.4	0.7 ± 0.3	1.4 ± 0.4	99.8 ± 0.1
3 hours	reg. self-play (ours)	0.2 ± 0.1	0.6 ± 0.4	99.6 ± 0.2	0.973 ± 0.005	1.6 ± 0.4	0.6 ± 0.2	1.2 ± 0.3	100.0 ± 0.0
30 hours	reg. self-play (ours)	0.0 ± 0.0	0.3 ± 0.2	99.7 ± 0.2	0.976 ± 0.005	1.4 ± 0.4	0.8 ± 0.3	1.1 ± 0.3	99.9 ± 0.0

229 4.2 The Role of Metadata in Generalization

230 The results in the previous section used the best policies across every category, trained within 50k
 231 scenarios and focused on scaling laws for human *demonstration* data. Next two demonstration data,
 232 a second and distinct source of data is scenario *metadata*: road graphs, initial agent positions, and
 233 velocities. Metadata is typically cheaper to obtain than human demonstrations and ships with most
 234 available driving datasets [23, 60]. Recent work has shown that regularized self-play RL grounded by
 235 target-city metadata can adapt driving policies to new cities [22]. A natural follow-up question is
 236 how much the diversity provided by metadata matters for training generalizable policies, which is
 237 what we explore here.

238 To do this, we train regularized and unregularized agents on nested subsets $\mathcal{M}_k$ with $|\mathcal{M}_k| \in$
 239 $\{10, 100, 1,000, 10,000, 50,000\}$ scenarios, holding the BC anchors τ^n and reward function r fixed.

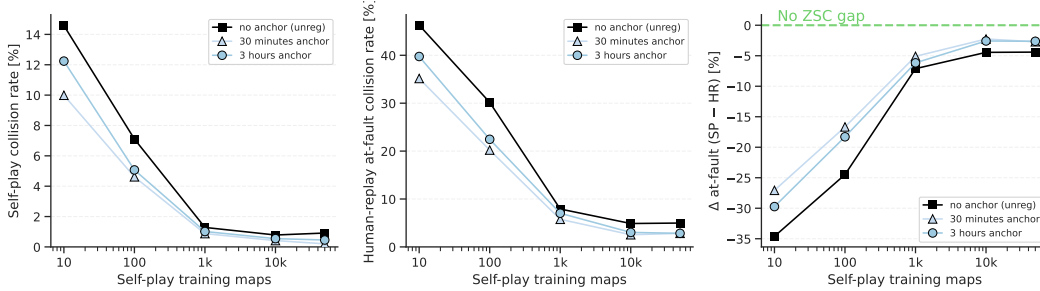


Figure 4: **Scaling laws for scenario metadata.** The unregularized self-play baseline is shown in black; shades of blue correspond to regularized policies trained with different BC anchors, with darker shades indicating more anchor data. *Left*: collision rate in self-play, where all agents are controlled by the same policy on a held-out validation set. *Center*: at-fault collision rate, the fraction of collisions caused by the controlled agent. This metric is more informative than raw collision rate, because replay agents frequently strike the SDC from behind, and such collisions are unlikely to occur in practice (See cartoon in Figure 2). *Right*: ZSC gap Δ_{ZSC} , the difference in the at-fault collision rate between the self-play and human-replay settings.

240 This isolates the effect of *scenario diversity*. Each policy is trained for 20 billion steps; Figure **TODO**
 241 confirms convergence on the training set.

242 To improve the signal of the comparison between policies, we evaluate on an *interactive* partition of the
 243 held-out set: scenarios filtered to contain dense multi-agent interactions such as merges, unprotected
 244 turns, and yielding, where coordination with other drivers is essential (details in Appendix D.1).
 245 This is done to compensate for the fact that many scenarios in the Waymo Open Dataset involve
 246 little interaction [21, 61], which compresses differences between policies regarding their ability to
 247 coordinate. Figure 4 and Table 2 report results on this partition.

248 **Scene diversity is key for generalizing to unseen maps and for zero-shot coordination with the**
 249 **human population.** Both vanilla and regularized self-play improve sharply with the amount of
 250 metadata. For vanilla self-play, the at-fault collision rate drops from 43.8% at 10 scenarios to 1%
 251 at 10k scenarios, and the human-replay collision rate falls from 40% to 4.1% over the same range.
 252 Regularized self-play follows the same trend and reaches much lower absolute values: with a 3-hour
 253 BC anchor, the at-fault collision rate drops from 40% at 10 scenarios to 2.4% at 10k scenarios. The
 254 ZSC gap closes in parallel, shrinking from a self-play score that far exceeds the human-replay score
 255 to near parity at 10k scenarios (self-play 0.971, human-replay 0.945 for the 3-hour anchor).

256 **Vanilla self-play stagnates in human coordination.** Map diversity yields substantial gains up
 257 to 10k scenarios; the HR score rises from 0.312 (10 maps) to 0.891 (10k maps), and the at-fault
 258 collision rate falls from 46.2% to 4.1%; but shows diminishing returns beyond that point. Scaling
 259 to 50k maps modestly improves the HR score to 0.908 yet does not reduce the at-fault rate, which
 260 remains at 4.9%, within noise of the 10k result. This pattern suggests that scene diversity alone may
 261 be insufficient to close the human-coordination gap.

262 **Regularized self-play reduces the human-coordination gap.** Adding a BC anchor improves
 263 human-coordination metrics across map scales. At 10k maps, a 10-minute anchor reduces the at-fault
 264 collision rate to 3.0% (HR score 0.916), and a 30-minute anchor to 2.4% (HR score 0.926), both
 265 below the unregularized baseline of 4.1%. Interestingly, a 3-hour anchor at the same map scale does
 266 not continue this trend (3.5%, HR score 0.906), indicating that the relationship between anchor size
 267 and performance is not monotone at fixed map diversity, though the reasons for this remain unclear.
 268 Combining larger map diversity with regularization generally yields the strongest results: at 50k
 269 maps, a 30-minute anchor achieves 2.6% at-fault (HR score 0.931), and 30 hours of anchor data gives
 270 the best observed result of 2.1% at-fault with an HR score of 0.949 and IDM score of 0.908.

Table 2: Interactive evaluation across all scaling checkpoints. All metrics are computed on the interactive validation subset; policies are rolled out in each or the 200 scenarios 10 times. Top-3 values per column are highlighted (**best** , **2nd** , **3rd**); best value additionally in bold. Gray marks the best unregularized value per column.

Self-play maps (metadata)	Anchor data (human demos)	Score		Collision rates			
		HR Score $\uparrow$	IDM Score $\uparrow$	IDM At-fault (%) $\downarrow$	HR At-fault (%) $\downarrow$	IDM Coll. (%) $\downarrow$	HR Coll. (%) $\downarrow$
10	0 (unreg.)	0.312 $\pm$ 0.010	0.296 $\pm$ 0.010	42.8 $\pm$ 1.1	46.2 $\pm$ 1.1	46.6 $\pm$ 1.1	50.1 $\pm$ 1.1
100	0 (unreg.)	0.598 $\pm$ 0.011	0.577 $\pm$ 0.011	28.9 $\pm$ 1.0	29.9 $\pm$ 1.0	34.6 $\pm$ 1.1	34.3 $\pm$ 1.0
1k	0 (unreg.)	0.868 $\pm$ 0.007	0.842 $\pm$ 0.008	5.8 $\pm$ 0.5	7.6 $\pm$ 0.6	10.1 $\pm$ 0.7	12.2 $\pm$ 0.7
10k	0 (unreg.)	0.891 $\pm$ 0.007	0.876 $\pm$ 0.007	3.2 $\pm$ 0.4	4.1 $\pm$ 0.4	9.0 $\pm$ 0.6	10.2 $\pm$ 0.7
50k	0 (unreg.)	0.908 $\pm$ 0.006	0.893 $\pm$ 0.007	3.8 $\pm$ 0.4	4.9 $\pm$ 0.5	7.6 $\pm$ 0.6	8.7 $\pm$ 0.6
10	30 minutes	0.425 $\pm$ 0.011	0.432 $\pm$ 0.011	33.1 $\pm$ 1.0	34.6 $\pm$ 1.1	36.6 $\pm$ 1.1	37.6 $\pm$ 1.1
10	3 hours	0.361 $\pm$ 0.011	0.371 $\pm$ 0.011	37.3 $\pm$ 1.1	39.6 $\pm$ 1.1	39.8 $\pm$ 1.1	43.2 $\pm$ 1.1
100	30 minutes	0.722 $\pm$ 0.010	0.661 $\pm$ 0.010	16.8 $\pm$ 0.8	18.0 $\pm$ 0.8	22.4 $\pm$ 0.9	23.6 $\pm$ 0.9
100	3 hours	0.658 $\pm$ 0.010	0.629 $\pm$ 0.011	21.8 $\pm$ 0.9	24.0 $\pm$ 0.9	25.5 $\pm$ 1.0	28.2 $\pm$ 1.0
1k	30 minutes	0.897 $\pm$ 0.007	0.858 $\pm$ 0.008	4.4 $\pm$ 0.5	5.9 $\pm$ 0.5	8.4 $\pm$ 0.6	9.2 $\pm$ 0.6
1k	3 hours	0.886 $\pm$ 0.007	0.866 $\pm$ 0.008	5.3 $\pm$ 0.5	7.0 $\pm$ 0.6	9.3 $\pm$ 0.6	10.2 $\pm$ 0.7
10k	10 minutes	0.916 $\pm$ 0.006	0.858 $\pm$ 0.008	3.1 $\pm$ 0.4	3.0 $\pm$ 0.4	8.3 $\pm$ 0.6	6.8 $\pm$ 0.6
10k	30 minutes	0.926 $\pm$ 0.006	0.892 $\pm$ 0.007	3.5 $\pm$ 0.4	2.4 $\pm$ 0.3	7.9 $\pm$ 0.6	7.1 $\pm$ 0.6
10k	3 hours	0.906 $\pm$ 0.006	0.873 $\pm$ 0.007	3.0 $\pm$ 0.4	3.5 $\pm$ 0.4	7.7 $\pm$ 0.6	7.9 $\pm$ 0.6
10k	30 hours	0.925 $\pm$ 0.006	0.904 $\pm$ 0.007	2.6 $\pm$ 0.4	3.5 $\pm$ 0.4	5.9 $\pm$ 0.5	6.0 $\pm$ 0.5
50k	10 minutes	0.923 $\pm$ 0.006	0.883 $\pm$ 0.007	3.1 $\pm$ 0.4	3.0 $\pm$ 0.4	7.4 $\pm$ 0.6	6.9 $\pm$ 0.6
50k	30 minutes	0.931 $\pm$ 0.006	0.890 $\pm$ 0.007	2.8 $\pm$ 0.4	2.6 $\pm$ 0.4	5.6 $\pm$ 0.5	6.0 $\pm$ 0.5
50k	3 hours	0.935 $\pm$ 0.005	0.890 $\pm$ 0.007	3.6 $\pm$ 0.4	2.8 $\pm$ 0.4	6.5 $\pm$ 0.5	5.2 $\pm$ 0.5
50k	30 hours	0.949 $\pm$ 0.005	0.908 $\pm$ 0.006	2.2 $\pm$ 0.3	2.1 $\pm$ 0.3	5.2 $\pm$ 0.5	4.2 $\pm$ 0.4

4.3 Behavior and Safety Analysis

Regularized agents exhibit lower-severity collisions. Collision rates, as reported in Sections 4.1 and 4.2, measure how often agents fail, but not how bad the failures are. This distinction matters when policies are deployed alongside humans. Following Waymo’s most recent safety report [62] (details in Appendix D.2), we quantify collision severity via the *change in velocity at impact* (Δv), a widely studied proxy for occupant injury risk. As shown in Table 3 and Figure 5, regularization improves both the frequency and the severity of failures. The mean per-event Δv drops from 2.09 m/s to 1.71 m/s, and the maximum observed impact velocity falls from 13.71 m/s to 8.09 m/s. The improvement is more apparent when we focus on the tail of collision events: 14.3% of unregularized collisions exceed 15 mph, the threshold above which serious injury risk rises substantially, compared to 7.5% for the regularized model. This is roughly a two-fold reduction. The survival curve in Figure 5 (right) shows the two groups are nearly indistinguishable at low Δv , with the gap opening sharply above 5 m/s and widening through the severe range. Human regularization thus produces agents that not only collide less often, but fail more gracefully: when a collision is unavoidable, the regularized policy arrives at impact with lower relative speed, yielding a severity profile that resembles more of a cautious human driver.

Table 3: Collision severity tail breakdown with human-replays in interactive held-out scenarios. *Events* shows the count and share of all collision events attributed to each group. Per-event Δv statistics and the fraction of events exceeding three injury-risk thresholds (1 mph: cosmetic; 5 mph: airbag-deployment floor; 15 mph: elevated serious-injury risk). Best value per column in **bold**; lower is better throughout.

Method	Events (at-fault coll. rate)	Mean Δv (m/s) $\downarrow$	Max Δv (m/s) $\downarrow$	> 1 mph (%) $\downarrow$	> 5 mph (%) $\downarrow$	> 15 mph (%) $\downarrow$
unregularized	91 (5.0%)	2.09	13.71	89.0	54.9	14.3
regularized	53 (2.8%)	1.71	8.09	90.6	54.7	7.5

Distributional realism via WOSAC. We analyze the Waymo Open Sim Agent Challenge (WOSAC) scores to compare the distributional realism of regularized self-play agents against vanilla self-play and SMART. Figure 6 reports the WOSAC realism meta-score alongside its three group metrics (kinematic, interactive, and map-based); Figure 18 breaks down all nine submetrics that together make up the meta-score.

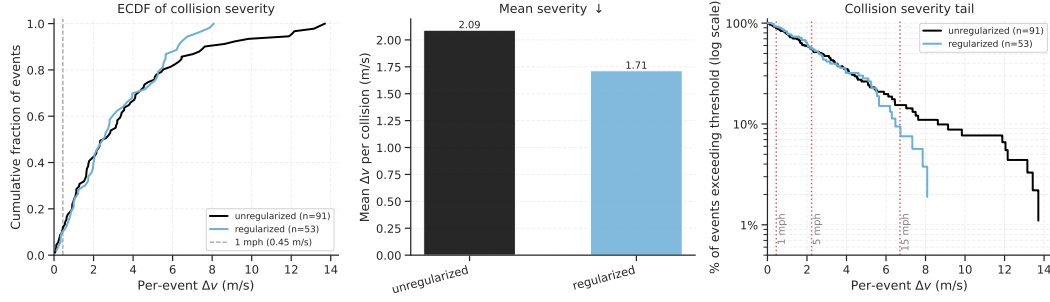


Figure 5: **Regularized self-play RL agents exhibit lower-severity at-fault collisions.** **Left:** empirical CDF of per-event Δv . The regularized model’s distribution is shifted left of the unregularized baseline at every quantile, with the gap widening through the moderate-severity range (2–5 m/s). The dashed line marks Waymo’s 1 mph (0.45 m/s) reporting threshold. **Center:** mean Δv per collision event, conditional on a collision occurring. Regularized collisions are on average 18% lower in severity (1.71 vs. 2.09 m/s). **Right:** fraction of collision events exceeding Δv (log scale). The two groups are nearly indistinguishable in the low-severity regime; the gap opens above 5 mph and roughly doubles by 15 mph (14.3% vs. 7.5%), confirming that regularization reduces serious impacts disproportionately relative to minor ones.

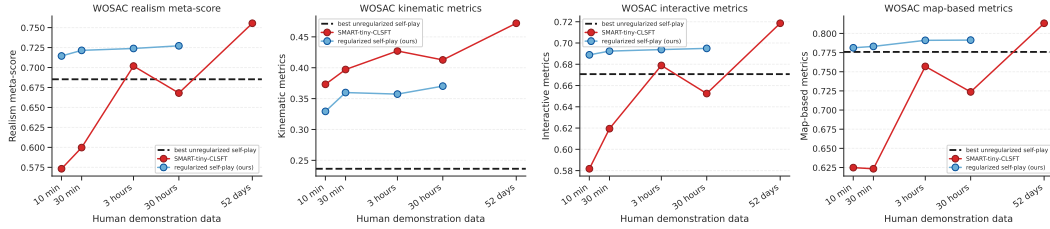


Figure 6: WOSAC scores

Vanilla self-play agents score 0.68, losing points primarily in kinematics (0.22) and the interactive category. In particular, they maintain less distance to neighboring agents compared to the human logs, and exhibit increased time-to-collision (TTC) rates, reflecting the absence of human-like spatial buffering. Regularized self-play agents reach a meta-score of 0.725, and this score is roughly stable across the full range of human data (30 minutes to 30 hours), suggesting that a small human data budget with the existing policies is sufficient to recover these realism gains. The remaining deficit appears in kinematic metrics (0.36), specifically linear speed, and in the distance-to-nearest-object submetric, indicating that while regularization meaningfully closes the gap to human behavior, some room for improvement remains. SMART at 52 days of training data achieves the highest WOSAC score of 0.75, outperforming regularized self-play on this metric despite underperforming on collision rate and task completion across all data bins.

We should note that while WOSAC is informative, the benchmark is not perfect. WOSAC measures how closely agent behavior matches the distributional statistics of human logs, but it is not obvious that higher WOSAC scores translate directly into safer or more capable driving [63].

Regularized agents exhibit more social driving behavior. To complement the quantitative results, we analyze a series of videos to study behavioral differences between regularized and unregularized agents. Representative videos are available at <https://sites.google.com/view/anonymous-human-like-autonomy/>.

The most salient qualitative difference is that regularized agents are more considerate of surrounding traffic: they maintain greater distance to neighboring agents, avoid cutting in, and wait for others at

intersections. This is consistent with the known tendency of goal-conditioned RL agents to reach their objective as efficiently as possible (due to discounting). Notably, this behavioral gap is largely invisible in the collision metrics, since most such incompatibilities do not result in failures. Yet they represent good examples of the type of behavior that would be unsafe and alien in real deployment alongside humans. The trajectory metrics in Table 4 offer a partial window into this effect: regularized agents achieve substantially lower longitudinal L2, lateral L2, and displacement errors, reflecting closer alignment with how humans would navigate the interactive traffic scenes.

Table 4: Main results comparing unregularized and regularized self-play at 50k training maps. Self-play score on 10k validation scenes; human-replay and IDM-replay metrics on 200 interactive validation scenes. Best value per column in **bold**.

Method	Self-play	Score		IDM	At-fault (%) ↓	Human-replay (interactive)		ADE ↓
		HR				Long. L2 ↓	Lat. L2 ↓	
Unregularized	0.986 ± 0.002	0.908 ± 0.006	0.893 ± 0.007	4.9 ± 0.5	13.327 ± 0.129	2.390 ± 0.148	14.074 ± 0.182	
Regularized (ours)	0.975 ± 0.002	0.931 ± 0.006	0.890 ± 0.007	2.6 ± 0.4	5.559 ± 0.077	1.274 ± 0.029	5.927 ± 0.076	

Regularization keeps RL policies close to human anchors. Figure 7 shows task completion and KL divergence to the anchor policy over training. Both regularized and unregularized agents converge to comparably effective strategies: they learn to navigate to their goals without colliding. Yet the action distributions underlying these strategies differ substantially. Unregularized self-play RL drifts freely through the space of effective policies and converges to a strategy far from human behavior; the KL divergence increases monotonically throughout training. Regularized self-play, by contrast, remains anchored. The agent is free to discover competent strategies, but the regularization penalty keeps those strategies close to the behavioral distribution induced by human demonstrations. This is the key idea of regularized self-play: we do not constrain what the agent achieves but only how it achieves it. The result is an agent that is both capable (solving the driving task through large-scale self-play) and compatible (staying within the distribution of strategies a human driver would recognize as natural).

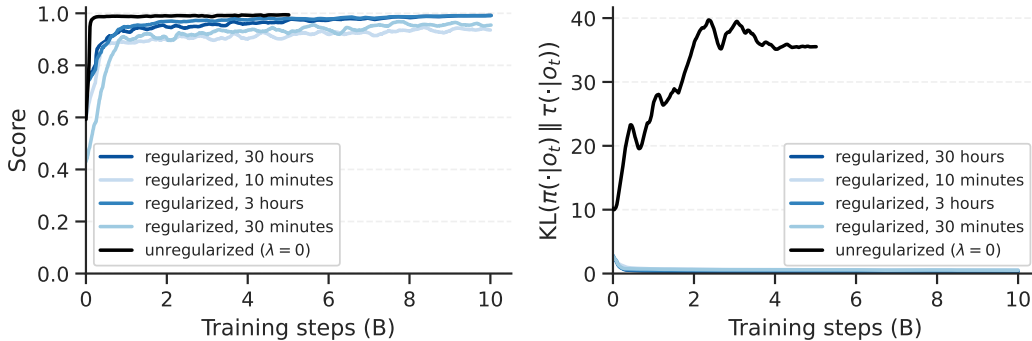


Figure 7: **TODO: Update with 20 Billion runs** Regularized self-play policies stay close to the anchor distribution while unregularized self-play diverges. Here we show that, while both unregularized (black) and regularized self-play policies converge to effective strategies (left), the action distribution underlying these strategies varies widely as indicated by the kl divergence between the observation-conditioned action distributions (right). The regularized self-play policies stay close to the anchors, whereas the unregularized policies diverge.

5 Conclusion & Limitations

We considered a series of experiments aimed at putting the mixing of human demonstration data with synthetic simulated experience on a more scientific footing. Our central finding is that a small amount of human data, roughly 30 minutes to 3 hours of demonstrations, suffices to train human-compatible

driving agents when used as a behavioral cloning anchor to regularize self-play RL against 60 years of synthetic experience. This is three orders of magnitude less than state-of-the-art imitation learning baselines, eliminates the need for reward shaping, and is achieved at a fraction of the training time. The broader implication is that, wherever simulation is cheap, human data may be best used **not** as the primary training signal but as a *lightweight anchor* that steers policies away from effective-but-alien equilibria.

This empirical result raises a deeper question that we have only touched the surface of: what is the *complementary value* of a bit of human data given a batch of synthetic experience? Can we quantify this? We can loosely intuit that self-play trains more robust agents because it covers a wider distribution of states, and indeed, our self-play agents learn from 20 billion transitions against 225 million for the IL baseline (Figure 1). But sheer count is a crude explanation; not all transitions are equally informative. Recent work on Epiplexity [64] takes a step toward formalizing this, conceptually defining the value of data as the amount of learnable structure it contains. Translating such notions into practical tools for selecting or filtering data remains open.

As for limitations, while regularized self-play improves on prior work in multiple dimensions, several aspects remain to be improved. First, our evaluation relies on human-replays and IDM as a proxy for deployment, and it is currently unclear to what extent performance in the self-play and human-replay settings *exactly translates* to behavior on the road; closing the sim-to-real gap remains future work. Second, while our best policy reaches a near-zero at-fault collision rate (0.6–0.8%), the overall collision rate is somewhat higher (1.4–2.0%), partly because replay agents can strike the SDC from behind in situations the policy cannot reasonably avoid. There is room for improvement in training agents that are both human-compatible and robust. However, what constitutes a fair collision-avoidance benchmark beyond at-fault heuristics is itself a difficult open question in both industry and academia [65], and we leave a more principled treatment of this to future work.

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544 A Simulation Environment and Design

545 A.1 World Initialization from Scenario Metadata

546 We use PufferDrive 2.0 for simulation and training [18]. PufferDrive is a batched simulator that runs
 547 many environments in parallel, reaching 400k steps per second (SPS) on an NVIDIA RTX 5090 GPU.
 548 We initialize environments using the Waymo Open Motion Dataset (WOMD) [23], which provides a
 549 large set of multi-agent traffic scenarios. Each scenario supplies the metadata we need: the roadgraph,
 550 a variable number of agents (cars, cyclists, and pedestrians), and other objects in the scene. This
 551 information is the output of a perception stack, so we operate directly on these clean features (in
 552 bounding-box world).

553 Each scenario is 9 seconds long and discretized into 90 steps. We take each logged agent’s initial
 554 position ($t = 0$) as its starting position in the scene, and its last valid logged position ($t = T$) as
 555 its goal, which lets us goal-condition the agents. The full Waymo training dataset contains 500k
 556 scenarios, but in this paper we use at most 50k of the randomly sampled scenarios. When constructing
 557 the environments, we randomly sample scenarios from WOMD until we hit a target number of agents
 558 (e.g., on an NVIDIA RTX 4080 with 16GB of memory, we keep adding environments until we reach
 559 1024 agents).

560 A.2 Observation Space

561 We take a decentralized approach and provide every agent with a partial view of the environment
 562 in a local coordinate frame. This is similar to the observation space of prior related works, such
 563 as GIGAFLow [5], and GPUDrive [4]. At each timestep, an agent receives the combination of
 564 three feature blocks: an ego block describing its own state, a partner block describing the $N_p = 31$
 565 closest other agents within a 50 m radius, and a road block describing up to $N_r = 128$ nearby
 566 road segments drawn from a 21×21 grid of $5 \text{ m} \times 5 \text{ m}$ cells centered on the agent. Missing slots
 567 (fewer partners or road segments than the maximum) are zero-padded. Tables 5, 6, and 7 list the
 568 features in each block. All positions and headings are expressed in the agent’s local frame, so the
 569 observation is invariant to the global pose of the scene. The total observation vector has dimension
 570 $11 + 7 \times 31 + 7 \times 128 = 1,124$.

Table 5: Ego features (14 values) for the delta-local dynamics model. Features 0–3 expose the sampled conditioning variables to the policy so it can modulate its behavior as a function of λ and the reward weights (Section 4.2). This was not used further in the paper; all values are kept at a fixed value.

Idx	Feature	Normalization	Description
0	λ	—	Human-regularization coefficient
1	r_{coll}	—	Sampled collision reward
2	r_{off}	—	Sampled off-road reward
3	r_{goal}	—	Sampled goal reward
4	Δx_{goal}	$\times 0.005$	Goal position (ego frame), longitudinal
5	Δy_{goal}	$\times 0.005$	Goal position (ego frame), lateral
6	signed speed	/ 100 m/s	Speed projected onto heading
7	vehicle width	/ 15 m	Ego bounding-box width
8	vehicle length	/ 30 m	Ego bounding-box length
9	collision flag	$\{0, 1\}$	1 if currently colliding
10	entity type	/ 3	Vehicle (1), pedestrian (2), cyclist (3)

Table 6: Partner features (7 values $\times$ 31 partners = 217 values). Partners are ordered by index and filtered to those within 50 m of the ego agent. All positions and headings are in the ego frame.

Idx	Feature	Normalization	Description
0	Δx	$\times 0.02$	Partner position, longitudinal
1	Δy	$\times 0.02$	Partner position, lateral
2	partner width	/ 15 m	Partner bounding-box width
3	partner length	/ 30 m	Partner bounding-box length
4	$\cos(\Delta\psi)$	—	Relative heading, cosine component
5	$\sin(\Delta\psi)$	—	Relative heading, sine component
6	partner signed speed	/ 100 m/s	Signed speed along partner’s heading

Table 7: Road-segment features (7 values $\times$ 128 segments = 896 values). Segments are drawn from a 21×21 grid of 5 m cells centered on the ego agent, and include road lanes, road lines, and road edges. Each segment is described by the midpoint, length, and orientation of a single polyline segment.

Idx	Feature	Normalization	Description
0	midpoint x	$\times 0.02$	Segment midpoint, longitudinal (ego frame)
1	midpoint y	$\times 0.02$	Segment midpoint, lateral (ego frame)
2	segment length	/ 100 m	Length of the polyline segment
3	segment width	/ 100 m	Fixed nominal width (0.1 m)
4	$\cos(\theta)$	—	Segment orientation in ego frame
5	$\sin(\theta)$	—	Segment orientation in ego frame
6	segment type	$\{0, 1, 2\}$	Road lane (0), road line (1), road edge (2)

571 A.3 Actions and Dynamics

572 We use a single dynamics model with a discretized action space for both the unregularized and
573 regularized agents.

574 **Delta-local dynamics with kinematic constraints.** The action is a triple $(\Delta x, \Delta y, \Delta\psi)$ in the
575 agent’s local frame at time t . Translation is rotated into the world frame and added to the position;
576 heading is updated directly:

$$x_{t+1} = x_t + \cos(\psi_t) \Delta x - \sin(\psi_t) \Delta y, \quad (3)$$

$$y_{t+1} = y_t + \sin(\psi_t) \Delta x + \cos(\psi_t) \Delta y, \quad (4)$$

$$\psi_{t+1} = \text{wrap}(\psi_t + \Delta\psi). \quad (5)$$

577 Velocity is reported as the world-frame displacement divided by $\Delta t = 0.1$ s. We bound each
578 component roughly based on realistic actions present in the human demonstrations, as shown in
579 Figure 8; specifically, we define $\Delta x \in [-3.5, 3.5]$ m, $\Delta y \in [-0.1, 0.1]$ m, and $\Delta\psi \in [-\pi/6, \pi/6]$.
580 Each of the three dimensions is binned independently into 51, 51, and 127 values, respectively. Figure
581 8 shows that the distributions for Δy and $\Delta\psi$ are roughly symmetric, whereas the distribution for
582 Δx is strongly asymmetric. This is expected, since most vehicles move forward and only a small
583 number of agents in the scenes drive in reverse (e.g., when parking).

584 Delta-local dynamics are kinematically unconstrained by default: the agent can translate laterally
585 without rotating, pivot in place, or instantaneously reverse its heading rate. To prevent impossible
586 behaviors, we apply two physics-based constraints to the action at each step. Each constraint clips the
587 action after the previous one has been applied, with the previously executed (post-constraint) values
588 used as the reference. The constraints are:

- 589 1. **Longitudinal acceleration bound.** The change in implied forward speed is clipped to
590 $\pm A_{\text{long,max}} \cdot \Delta t$, where $A_{\text{long,max}} = 8 \text{ m/s}^2$. This caps acceleration and braking.

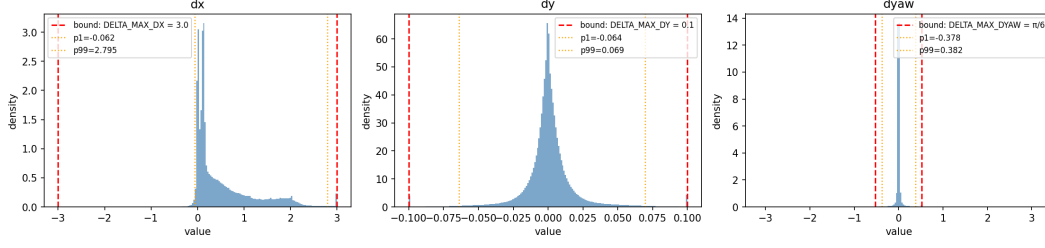


Figure 8: Discretized delta-local action space for each component (Δx , Δy , $\Delta \psi$). Histograms show the empirical density (blue) of 10,996,751 valid action timesteps recovered from expert trajectories across 10,000 maps. Yellow lines mark the 1st and 99th percentiles of the data; red lines mark the action-space bounds (± 3.5 m, ± 0.1 m, $\pm \pi/6$ rad). Each dimension is binned independently into 512 values. The bounds were chosen to respect natural movements in the data: 0.00% of Δx and Δy samples fall outside their bounds, and 0.71% of $\Delta \psi$ samples fall outside $\pm \pi/6$.

2. **Lateral motion envelope.** Lateral displacement is bounded by $|\Delta y| \leq |\Delta x| \cdot \tan(\delta_{\max})$, where $\delta_{\max} = 0.7$ rad is the maximum effective steering angle. This eliminates lateral sliding and side-shimmy at low forward speed.

These physical constraints prevent kinematically implausible actions; they do not encode any preference over driving style and are independent of the human anchor.

A.4 Collecting Human Demonstrations

The behavioral cloning (BC) anchor is trained on observation–action pairs (o_t, a_t) . We therefore need actions that (i) live in the simulator’s action space and (ii) reproduce the logged motion when applied through the simulator’s dynamics. We construct the dataset in two steps. Figure 10 shows three examples of this process in the simulator.

Step 1: Inferring actions from the logs. For each timestep t , we invert the delta-local dynamics to recover the action that produced the next logged state. Projecting the world-frame displacement into the agent’s local frame at t gives:

$$\Delta x_t = \cos(\psi_t)(x_{t+1} - x_t) + \sin(\psi_t)(y_{t+1} - y_t), \quad (6)$$

$$\Delta y_t = -\sin(\psi_t)(x_{t+1} - x_t) + \cos(\psi_t)(y_{t+1} - y_t), \quad (7)$$

$$\Delta \psi_t = \text{wrap}(\psi_{t+1} - \psi_t). \quad (8)$$

Each triple is clipped to the action bounds and snapped to the nearest discrete bin. Timesteps where either t or $t + 1$ is flagged invalid in the log are marked as missing and excluded from training.

Step 2: Replaying actions through the simulator. To produce observations, we replay the inferred action sequence through the simulator and record the observation at every resulting state. The BC anchor is then trained on the resulting (simulator observation, inferred action) pairs. Discretization introduces a small error that grows inversely with bin size (details below); to prevent its accumulation, we instead *teleport* agents to each successive state rather than stepping them forward with the inferred actions. We note that stepping agents directly is also viable when using larger action spaces, where the discretization error is smaller.

Effect of discretization on performance. Figure 9 and Table 8 quantify the cost of discretization. Continuous actions reproduce the logged trajectory almost exactly (ADE 0.001 m), confirming that the delta-local dynamics and kinematic constraints are themselves well-posed. Discretizing into 512 bins per dimension introduces a quantization floor of ADE 0.097 m, which is roughly two orders of magnitude larger, but is still very close to the original trajectory. Off-road and collision rates increase

modestly under discretization (1.2% vs. 0.8% off-road, 0.4% vs. 0.0% collision), reflecting the rare cases where snapping to the nearest bin pushes the SDC just outside a road edge or into a static neighbor; both representations complete the route in 100% of scenarios.

Table 8: Inferred-expert-action quality for the delta-local dynamics model. Comparison of discrete (bin-quantized) vs continuous (direct float) expert actions. Aggregated over 10,240 pooled samples. Values are mean $\pm$ SE.

Action type	Route prog. (%) $\uparrow$	Coll. (%) $\downarrow$	Off-road (%) $\downarrow$	ADE (m) $\downarrow$	Lateral L2 (m) $\downarrow$	Longitudinal L2 (m) $\downarrow$
discrete	100.0	0.4 ± 0.1	1.2 ± 0.2	0.097 ± 0.002	0.096 ± 0.002	0.004 ± 0.000
continuous	100.0	0.0	0.8 ± 0.1	0.001 ± 0.000	0.001 ± 0.000	0.001 ± 0.000

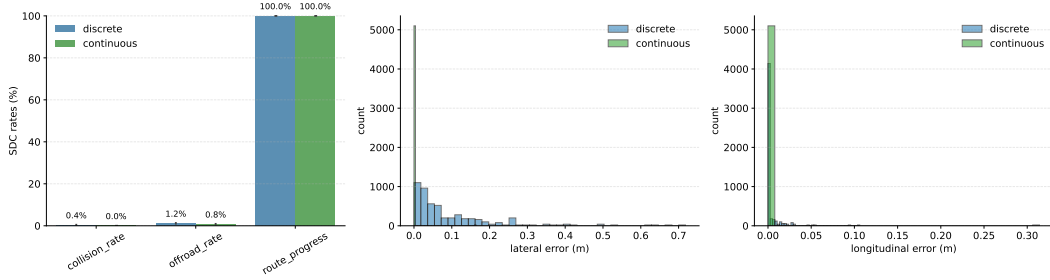


Figure 9: Effect of action discretization on inferred-expert-action quality. We replay each agent’s logged trajectory through the simulator using actions inferred from the logs, comparing discrete (bin-quantized, blue) and continuous (direct float, green) action representations. **Left:** SDC rates aggregated across 10,240 pooled samples; both representations complete the route in 100% of scenarios, but discretization induces modestly higher off-road and collision rates. **Center, right:** distributions of per-trajectory lateral and longitudinal L2 error to the logged pose. Continuous actions reproduce the log almost exactly (errors concentrated near zero), while discrete actions exhibit a small but consistent quantization floor of ~ 0.1 m laterally. Error bars on the bar plot denote standard error.

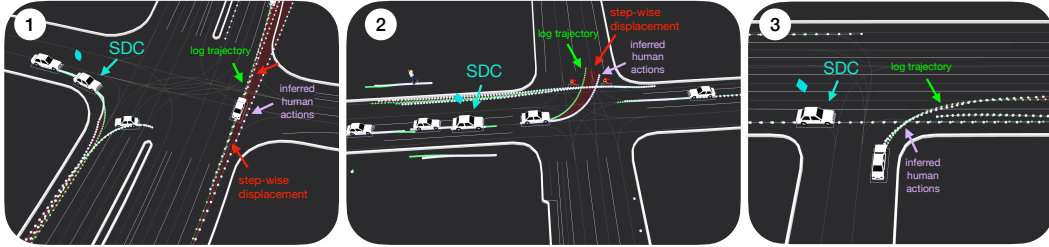


Figure 10: Three annotated example scenarios illustrating the human demonstration data collection process. The self-driving car (SDC), marked in cyan, is the Waymo vehicle whose human-driven trajectory we use as the demonstration. Logged trajectories are shown in green; purple trajectories show the result of stepping each agent through the simulator under the inferred delta-local actions. We select only the SDC trajectory because it is typically the cleanest demonstration in the scene; the visualized step-wise displacement illustrates a few low-quality (high-ADE) log trajectories that would otherwise contaminate the anchor.

621 A.5 Reward Function

622 We use a sparse reward: $r^i = +1$ if agent i reaches its goal within $\delta = 2$ meters before the episode
623 ends, -1 on collision or going off-road, and 0 otherwise. We deliberately omit dense shaping terms
624 so that safe and human-compatible behaviors can emerge from regularization.

Table 9: BC anchor evaluation. Open-loop metrics on the held-out validation set; closed-loop metrics averaged over validation scenes. Within-5-bin accuracy is the average of Δx , Δy , Δy_{aw} accuracies at the final training step.

Human data (h)	Acc. (%)	Open-loop		Closed-loop self-play		Closed-loop human-replay (SDC only)	
		Acc. ± 5 bins (%)	Loss	Route prog.	Score	Route prog.	Score
0.2	23.4	72.4	15.677	0.720 ± 0.012	0.215 ± 0.013	0.765 ± 0.007	0.242 ± 0.009
0.5	36.1	87.3	5.269	0.719 ± 0.011	0.277 ± 0.014	0.800 ± 0.006	0.371 ± 0.011
3.0	48.2	92.6	1.641	0.835 ± 0.010	0.502 ± 0.017	0.842 ± 0.006	0.538 ± 0.011
30.0	52.8	94.9	1.266	0.932 ± 0.007	0.685 ± 0.016	0.873 ± 0.006	0.666 ± 0.010

B Training

B.1 Behavioral Cloning Anchor Policies

Each anchor τ_n is trained by minimizing the negative log-likelihood of the logged actions under the factorized discrete action distribution described in Appendix A.3. We extract observation, action tuples through the procedure described in Appendix A.4. Note that we use only the SDC trajectory from each scenario for training, as it is the highest-quality demonstration. Since other agents are reconstructed from the perception stack, they exhibit more noise. Moreover, we have no guarantees about the driving quality of the surrounding humans. Since we obtain one trajectory per scene, each scenario contributes roughly 9 seconds of human data. Although these trajectories were collected in Waymo vehicles, they reflect manual human driving by an expert driver behind the wheel [23].

We train with Adam at a learning rate of 10^{-4} and a batch size of 2048 for up to 5000 epochs, with early stopping on the held-out validation loss after 100 epochs without improvement. Table 9 reports open- and closed-loop metrics for each anchor on 10,000 held-out validation scenarios. Figure 12 shows the 5-bin validation accuracy for each action head over training; from only 30 minutes of data, validation accuracy converges to between 80% and 90%. This metric is informative since there are 256 bins per action head, so the step sizes are very small.

Figures 14 and 13 compare the learned action distributions against the empirical distribution of the logged actions, for anchors trained on 30 minutes and 30 hours of data, respectively; in both cases, the learned distributions match the data reasonably well.

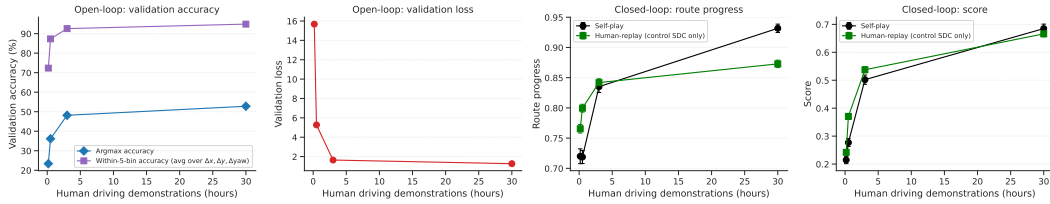


Figure 11: Open- and closed-loop performance of the anchor BC policies as a function of human demonstration data. Left: The final real (blue) and within 5 bin accuracy (purple) accuracy on 10,000 held-out validation scenarios. Right: Final validation loss. Right; Route progress; Right Score.

B.2 Self-Play Reinforcement Learning

B.2.1 Regularized Self-play RL Objective

Let π_θ denote the RL policy with parameters θ , and let τ_n be the fixed BC anchor trained on n scenarios.

Sampling. At the start of each episode, every agent i is assigned a regularization weight

$$\lambda_i \sim \text{Uniform}(0, 1), \quad (9)$$

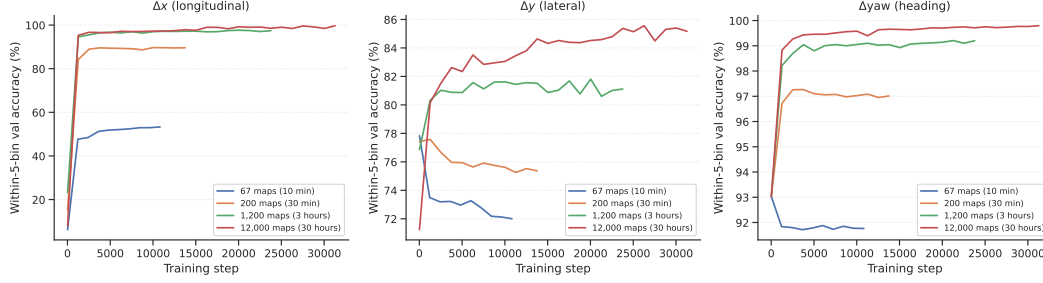


Figure 12: Training curves for the anchor policies. Each panel shows within-5-bin validation accuracy on a held-out set of scenarios for one action component (Δx , Δy , $\Delta\psi$). Curves terminate at different step counts because training stops once validation accuracy plateaus (no improvement for 100 consecutive epochs).

held fixed for the duration of the episode and resampled on reset. In our experiments, we give all agents the same regularization weight, $\lambda = 0.075$, which is held fixed during training and inference.

Regularization loss. We regularize π_θ toward τ_n by penalizing the KL divergence between the two policies on states visited during the rollout. For a minibatch of M transitions $\{(o_j, \lambda_j)\}_{j=1}^M$:

$$\mathcal{L}_{\text{reg}}(\theta) = \frac{1}{M} \sum_{j=1}^M \lambda_j D_{\text{KL}}(\tau_n(\cdot | o_j) \parallel \pi_\theta(\cdot | o_j, \lambda_j)), \quad (10)$$

where

$$D_{\text{KL}}(\tau_n \parallel \pi_\theta) = \sum_a \tau_n(a | o) [\log \tau_n(a | o) - \log \pi_\theta(a | o, \lambda)]. \quad (11)$$

Transitions with larger λ_j are pulled more strongly toward the anchor; when $\lambda_j = 0$ the regularization vanishes and the policy learns from the PPO objective alone. Because λ_j is part of the observation, π_θ learns to modulate its behavior as a continuous function of λ .

Full training objective. The standard PPO objective combines a clipped surrogate policy-gradient loss $\mathcal{L}_{\text{pg}}$, a value-function loss $\mathcal{L}_v$, and an entropy bonus H . The full training objective is

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{pg}} + c_v \mathcal{L}_v - c_H H + \mathcal{L}_{\text{reg}}, \quad (12)$$

where c_v and c_H are the value and entropy coefficients. No additional scalar multiplies $\mathcal{L}_{\text{reg}}$: the per-sample weighting by λ_j already controls the regularization strength.

Inference. In principle, λ can be set to any value in $[0, 1]$ to control the degree of human-likeness: $\lambda = 1$ would produce the most human-like behavior, and $\lambda = 0$ recovers the unregularized self-play policy.

B.2.2 Effect of Regularization on the Policy Gradient

Consider the gradient of the regularization loss with respect to θ .

Per-sample gradient. For a single rollout transition with observation o and conditioning value λ , the regularization loss contributes

$$\nabla_\theta \lambda D_{\text{KL}}(\tau_n(\cdot | o) \parallel \pi_\theta(\cdot | o, \lambda)) = -\lambda \sum_a \tau_n(a | o) \nabla_\theta \log \pi_\theta(a | o, \lambda). \quad (13)$$

This is a weighted behavioral cloning gradient: each action a is reinforced in proportion to the anchor’s probability $\tau_n(a | o)$, scaled by λ . When λ is large, the gradient strongly pushes π_θ toward the anchor; when $\lambda = 0$, the regularization term vanishes and only the PPO objective drives learning.

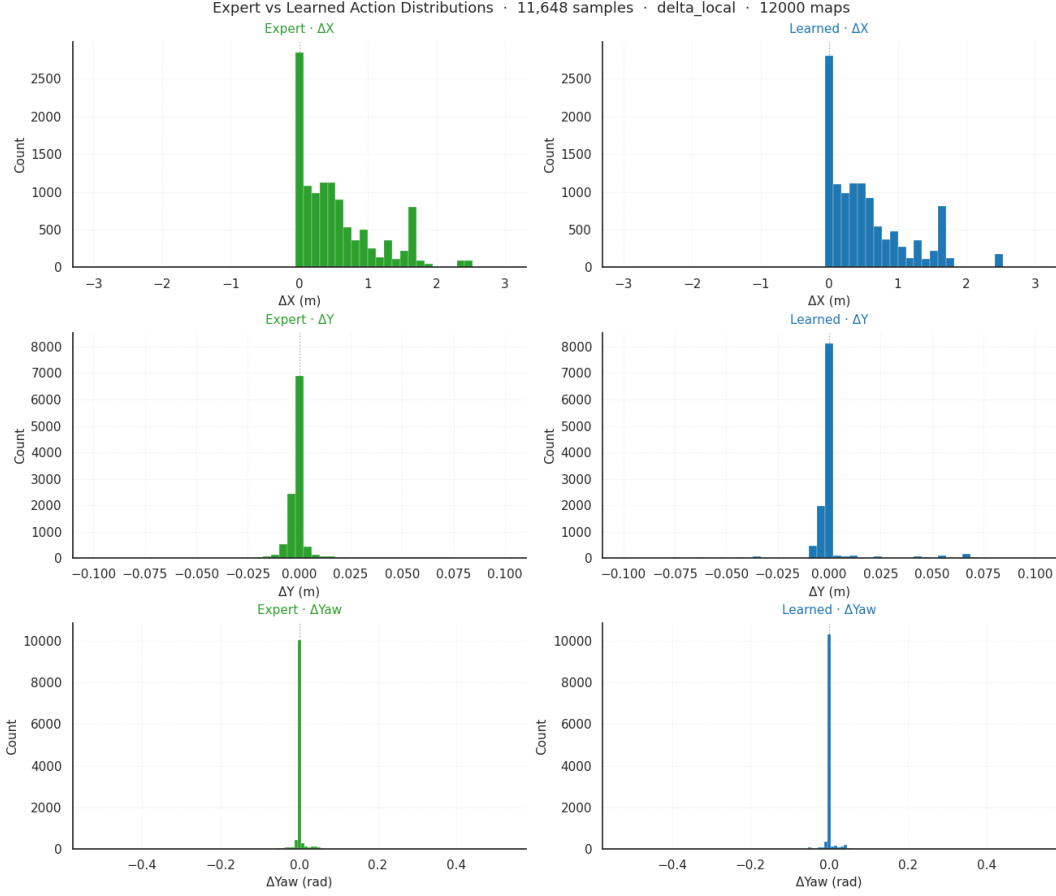


Figure 13: Example of actual vs. learned distributions - for 12k maps (30 hours)

671 **Interaction with the PPO gradients.** The PPO surrogate loss produces a gradient of the form

$$\nabla_{\theta} \mathcal{L}_{pg} \propto -\hat{A}(o, a) \nabla_{\theta} \log \pi_{\theta}(a | o, \lambda), \quad (14)$$

672 where $\hat{A}$ is the estimated advantage and a is the action taken during the rollout. The entropy bonus
673 contributes

$$-c_H \nabla_{\theta} H[\pi_{\theta}] = c_H \sum_a (1 + \log \pi_{\theta}(a | o, \lambda)) \nabla_{\theta} \log \pi_{\theta}(a | o, \lambda). \quad (15)$$

674 Combining all three terms gives

$$\underbrace{-\hat{A}(o, a) \nabla_{\theta} \log \pi_{\theta}(a | o, \lambda)}_{\text{policy gradient}} \quad (16)$$

$$+ \underbrace{c_H \sum_{a'} (1 + \log \pi_{\theta}(a' | o, \lambda)) \nabla_{\theta} \log \pi_{\theta}(a' | o, \lambda)}_{\text{entropy bonus}} \quad (17)$$

$$- \underbrace{\lambda \sum_{a'} \tau_n(a' | o) \nabla_{\theta} \log \pi_{\theta}(a' | o, \lambda)}_{\text{KL regularization}}. \quad (18)$$

675 **Entropy bonus vs. KL regularization.** The entropy bonus and the KL term are structurally similar:
676 both sum over the full action space and both act independently of the collected rewards. However,
677 they pull the policy in different directions. The entropy bonus weights each action by $1 + \log \pi_{\theta}(a)$,

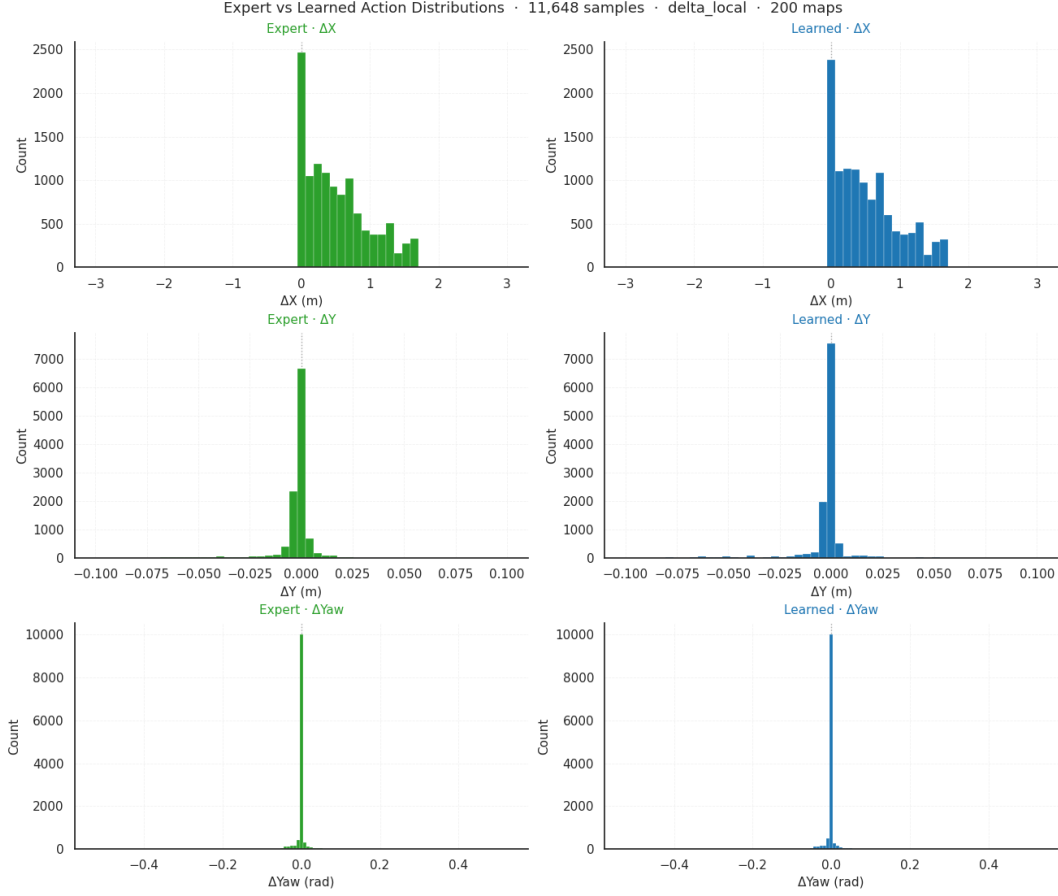


Figure 14: Example of actual vs. learned distributions - for 200 maps (30 min)

678 pushing probability mass toward the uniform distribution — an *untargeted* form of exploration. The
679 KL term weights each action by $\tau_n(a)$, pushing mass toward the anchor’s preferred actions — a
680 *targeted* form of regularization that steers the policy toward a reference distribution learned from
681 human data.

682 B.2.3 Baseline

683 We train the unregularized self-play baselines ($\lambda = 0$ throughout training) for 20 billion steps each
684 on $\{10, 100, 1,000, 10,000, 50,000\}$ maps, which is sufficient for convergence.

685 B.2.4 Data Filtering

686 We train only on transitions from agents that have not yet reached their goal. Once an agent is within
687 δ of its goal, it is removed from the scene and its remaining transitions are excluded from training.

688 B.2.5 Hyperparameters

689 We use PufferLib [17] for training. Hyperparameters are listed in Table 10. We use the native Protein
690 software for sweeps.

Table 10: PPO training hyperparameters.

Architecture		Training		Environment & Rewards	
Input size	64	Total timesteps	10B	Number of agents	1,024
Hidden size	256	Batch size	524,288	Number of workers	16
RNN type	LSTM	Minibatch size	32,768	Episode length	91 steps
RNN input size	256	Rollout horizon	32	Timestep Δt	0.1 s
RNN hidden size	256	Update epochs	1	Goal radius	2.5 m
		Learning rate	4.26×10^{-3}	Action space	Discrete
		LR schedule	Linear annealing	Dynamics model	Delta-local
		Adam β_1	0.9	Goal reward	+1.0
		Adam β_2	0.999	Collision penalty	-1.0
		Adam ϵ	10^{-8}	Off-road penalty	-1.0
		Clip coefficient	0.2		
		Entropy coefficient	0.001		
		VF coefficient	2.0		
		VF clip	0.2		
		GAE λ	0.95		
		Discount γ	0.99		
		Max gradient norm	1.0		
		Priority α	0.85		
		Priority β_0	0.85		
		V-trace c clip	1.0		
		V-trace ρ clip	1.0		
		Optimizer	Adam		
		Seed	42		

B.3 SMART Model Training

We trained SMART models in CATK on subsets of the Waymo Open Motion Dataset (WOMD). WOMD motion shards were preprocessed into CATK’s per-scenario cached format, and all training subsets were built from these cached scenario files. Our final local runs used `smart_mini_3M` with vehicle-only supervision on deterministic subsets of 67, 200, 1200, and 12000 scenarios. In the subset construction scripts we used, scenarios are sorted by cached scenario filename in lexicographic order. Vehicle-only supervision means that only vehicle agents contributed to the training loss, while pedestrians and cyclists were still present in the scene and available as context to the model. The local models were trained with CATK’s `pre_bc` configuration on a single GPU for 64 epochs with batch size 8. These runs used standard supervised behavioral cloning and were not further fine-tuned with CAT-K / CLSFT. We additionally compare against two author-provided checkpoints: a behavioral-cloning checkpoint (`pre_bc_E31.ckpt`) and a closed-loop supervised fine-tuned checkpoint (`clsft_E9.ckpt`). For downstream evaluation, we exported predictions as `.pkl` files on the 10k validation scenes from the `pufferdrive_womd_subsets` dataset. We used two export modes: an all-agents mode and a planning mode in which only the SDC is controlled by the model and all other agents are replayed from ground truth. We re-exported all model and export-mode combinations with 32 rollouts for multimodal evaluation. We verified that the scenario-ID overlap between each local training subset (67, 200, 1200, and 12000) and the evaluation set is zero.

C Neural Network Architecture

Both the BC anchor and the RL policy share the same multi-modal encoder structure. The flattened environment observation vector is first unpacked into its modalities: ego state, partner agents, and road segments. Each modality is processed by a two-layer MLP with ReLU activation and layer normalisation between the two linear layers. Partner and road embeddings are then aggregated across objects via max-pooling, producing one vector per stream. The three pooled vectors are concatenated and passed through a shared two-layer MLP (Linear $\rightarrow$ ReLU $\rightarrow$ Linear) to produce the final embedding. Separate linear heads decode this embedding into logits over each action dimension; a separate linear head with unit output produces the value estimate. The two architectures differ in width and in the presence of recurrence:

- **BC anchor.** Per-stream MLP width 128, shared MLP $3 \times 128 \rightarrow 512 \rightarrow 512$. No recurrence. Actor heads are linear projections from the 512-dimensional embedding. Total: 776,190 parameters.

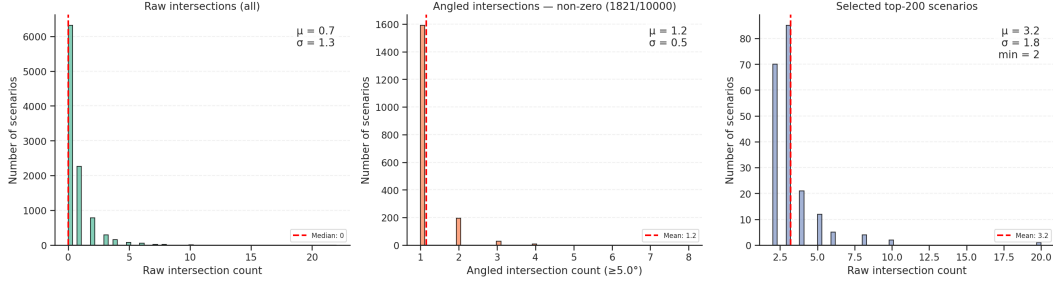


Figure 15: Distribution of SDC trajectory intersection counts. **Left:** raw intersection counts across all 50k scenarios. **Center:** angled intersections (non-zero only). **Right:** distribution within the selected top-200 subset.

- **RL policy.** Per-stream MLP width 64, shared MLP $3 \times 64 \rightarrow 256 \rightarrow 256$. The 256-dimensional embedding is passed through a single-layer LSTM with input size 256 and hidden size 256 (PufferLib LSTMWrapper). Actor and critic heads are linear projections from the 256-dimensional LSTM output. Total: **TODO**.

Road segment features include a categorical type field that is replaced by a 7-class one-hot vector before encoding, expanding the road feature dimension from d_{road} to $d_{\text{road}} + 6$.

D Evaluation

D.1 Filtering the Waymo Dataset for Interactive SDC Scenarios

Most scenarios in the Waymo Open Motion Dataset (WOMD) involve the self-driving car (SDC) traveling without meaningful interaction with other agents—the SDC reaches its destination without requiring coordination or yielding. To increase the signal in our human-replay evaluation, we filter the dataset for scenarios in which the SDC trajectory intersects with other agents’ trajectories, indicating situations that require coordination, such as merging, yielding, or navigating busy intersections.

We score each scenario by counting the number of segment-level intersections between the SDC trajectory and all other agent trajectories, optionally filtering crossings that meet a minimum acute-angle threshold (to exclude near-parallel overlaps, such as lane changes). From a pool of 10,000 held-out validation scenarios, we rank by intersection count and select the top 200 most interactive scenes. Figure 15 shows the resulting intersection count distributions across the full dataset and the selected subset, and Figure 16 shows nine representative examples from the selected set.

D.2 Metrics

We report the following metrics across all experiments. Unless noted, all metrics are computed per active agent per episode and averaged across agents and scenarios.

Score. The primary metric. An agent scores 1 if it reaches its goal without any collision or off-road event during the episode, and 0 otherwise. It jointly captures all failure modes and is the main basis for comparing conditions.

Completion rate. The fraction of agents that reach their goal position (within $\delta = 2$ meters) before episode end, regardless of whether a collision or off-road event occurred.

Collision rate. The fraction of episodes in which the agent is involved in at least one collision with another vehicle.

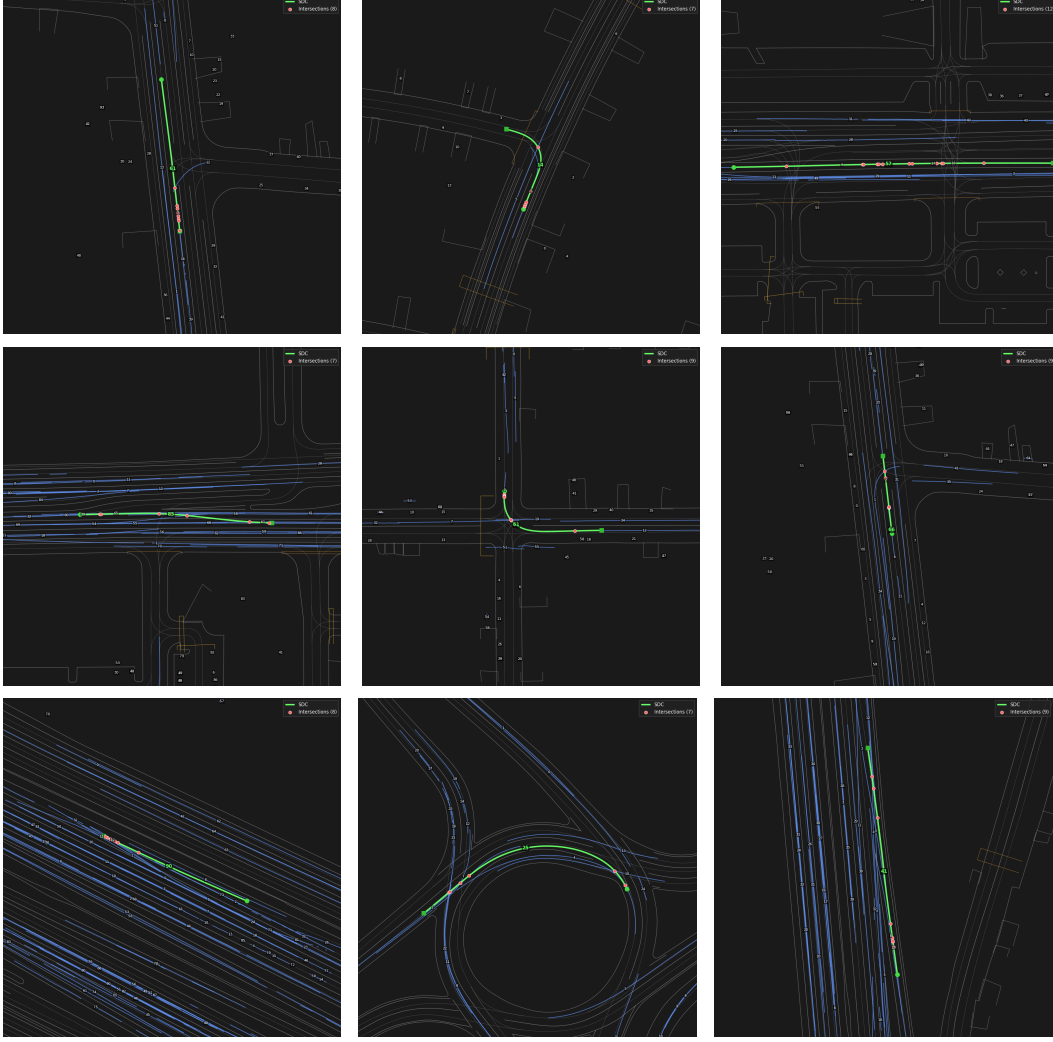


Figure 16: Nine example scenarios from the selected interactive subset. The SDC trajectory is shown in green, other agents in blue, and trajectory intersection points with other logs in red.

751 **At-fault collision rate.** A subset of the collision rate taken from NAVSIM [58]. A collision is
 752 attributed to an agent if (i) the other vehicle is in front of the agent at the time of impact, and (ii) the
 753 agent’s velocity vector points toward the other vehicle. This filters out collisions in which the agent
 754 was rear-ended or struck laterally by an inattentive partner.

755 **Rear collision rate.** The complement of at-fault: a collision is flagged as a rear collision for agent i
 756 if the colliding partner satisfies the at-fault conditions relative to i . A high rear collision rate indicates
 757 that the agent is being struck from behind, which is a signal that it is not keeping pace with traffic
 758 flow.

759 **Collision severity (Δv).** Beyond the binary collision indicator, we measure the severity of each
 760 at-fault collision event using the change in velocity (Δv) imparted to the agent at impact. Following
 761 the impulse-momentum formulation used in [62], the Delta-V of agent i in a collision with partner j
 762 is

$$\Delta v_i = \frac{m_j}{m_i + m_j} (1 + e) (\vec{v}_j - \vec{v}_i) \cdot \hat{n}, \quad (19)$$

763 where $\hat{n}$ is the unit collision normal (taken as the vector from agent i ’s center to agent j ’s center at
 764 impact), $e = 0.1$ is the coefficient of restitution for vehicle-to-vehicle crashes, and the dot product

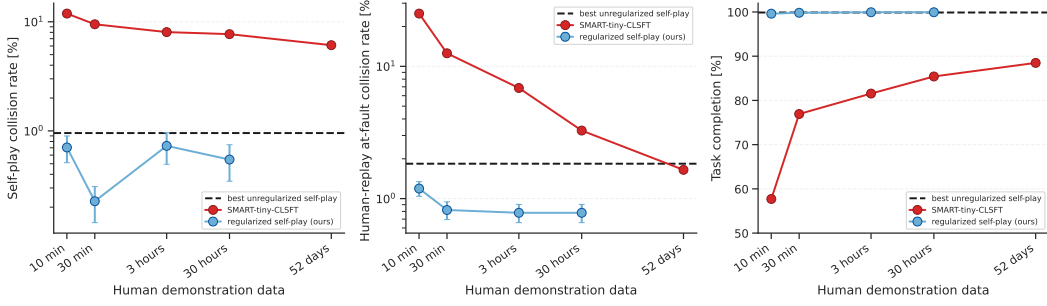


Figure 17: Scaling laws for human demonstration data; Same as Figure 3 but with the collision rates on a log scale.

is clipped at zero to ignore separating velocities. Masses are proxied from bounding-box footprint for vehicles (anchored at 1500 kg for a 4.5 m $\times$ 1.8 m reference sedan) and fixed for vulnerable road users (75 kg for pedestrians, 90 kg for cyclists). Δv is one of the strongest predictors of injury risk in vehicle-to-vehicle crashes [62] and lets us distinguish low-impact contacts (e.g. parking-lot taps) from high-energy collisions even when the binary collision rate is identical. We report two summaries: the mean Δv conditional on collision, and the fraction of collisions with $\Delta v < 1$ mph (0.447 m/s), the latter is chosen for direct comparability with Waymo’s reporting.

Off-road rate. The fraction of episodes in which the agent crosses a road edge boundary, detected by checking for intersection between the agent bounding box and any road edge polyline.

Route progress ratio. Following [66], we measure how far along its expert reference trajectory each agent travels. At each timestep t , we find the closest point $x(t)$ on the agent’s logged trajectory and compute its arc-length distance $d_{x(t)}$ from the start of the path. The route progress ratio is

$$\rho = \frac{d_{x(t)} - d_p}{d_q - d_p}, \quad (20)$$

where d_p and d_q are the arc-length distances to the initial and final positions of the logged trajectory, respectively. A value of $\rho = 1$ means the agent reached its destination; $\rho > 1$ is possible if the agent overshoots. For agents that reach their goal under GOAL_REMOVE termination, we set $\rho = 1$ directly, since their position is invalidated upon removal. For all other agents, ρ is computed from the agent’s final position at episode end.

Lateral deviation. At each timestep t , we compute the Euclidean distance from the agent’s current position to the nearest point on its expert reference trajectory. We average this distance over all timesteps for which the agent is alive, yielding a mean lateral deviation in meters. This measures how far the agent drifts sideways from its intended route; a perfect replay would achieve zero lateral deviation.

ZSC gap. The difference in score between the self-play and human-replay evaluation conditions:

$$\Delta_{\text{ZSC}} = \text{score}_{\text{self-play}} - \text{score}_{\text{human-replay}}. \quad (21)$$

A value of zero indicates that the policy transfers without degradation to the human population.

E Additional Results

E.1 Human demonstration data

E.2 Convergence and Performance on the Training Dataset

Every policy is trained for 5 billion steps. Below, we plot the performance on the full training dataset as a sanity check for convergence.

E.3 Human-Replay Results Comparison

E.4 Waymo Open Sim Agent Challenge

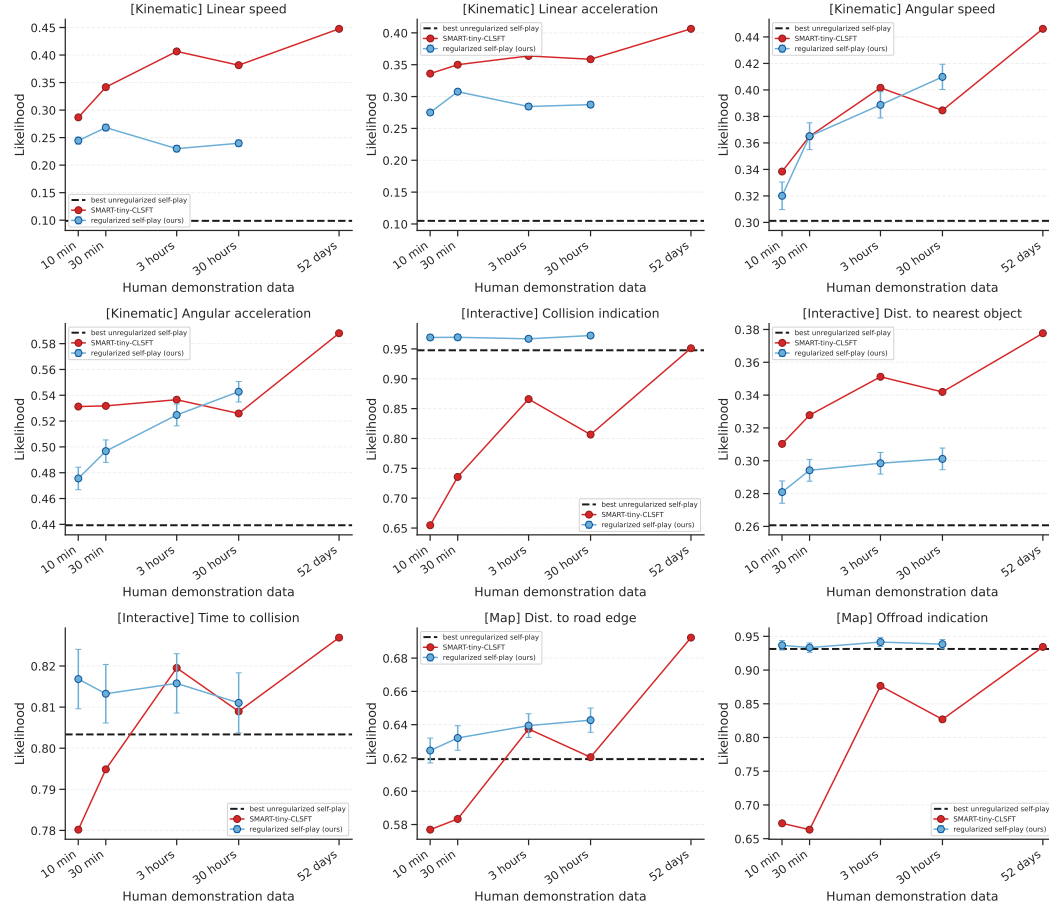


Figure 18: Submetrics

F Mapping Agent Experience to Human Time

We train self-play RL agents on 20 billion transitions. Since Waymo scenarios are discretized at 10 Hz, each transition (o_t, a_t) corresponds to 0.1 seconds of real time, placing the total training experience at approximately 31.68 years of driving.

For comparison, SMART [42] was trained on the full Waymo Open Motion Dataset, which contains 500,000 training scenarios. Each scenario contributes roughly 90 transitions of SDC trajectory data at 10 Hz, amounting to approximately 45 million transitions in total. The open-sourced SMART-CLSFT

803 checkpoint was trained on all agents in each scene rather than the SDC alone; assuming an average of
804 5 agents per scenario, this corresponds to roughly 225 million transitions.

805 Our own checkpoints are trained on subsets of 67, 200, 1,200, and 12,000 maps, each contributing
806 approximately 90 transitions per scenario.